

Understanding the Limits of **Single-Cell Foundation Models** on Downstream Tasks

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scRNA
GPT
LLM

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Motivation

Why evaluate single-cell foundation models?

The Foundation Model Revolution

Transformer architectures have transformed scRNA-seq analysis
Large-scale pretraining on millions of transcriptomes
Promise of transferable representations for
Multiple downstream tasks
Multiple competing models: Geneformer, scGPT, scFoundation

The Evaluation Gap

Reported performances are difficult to compare
Heterogeneous datasets & preprocessing pipelines
Different fine-tuning strategies across papers
Unclear when pretraining provides real advantages

Key Research Questions

- Q1 What biological information is robustly encoded during pretraining?
- Q2 Under what conditions do these representations fail to generalize?
- Q3 How do frozen representations compare to fine-tuned models?

Background

Single-cell transcriptomics & foundation models

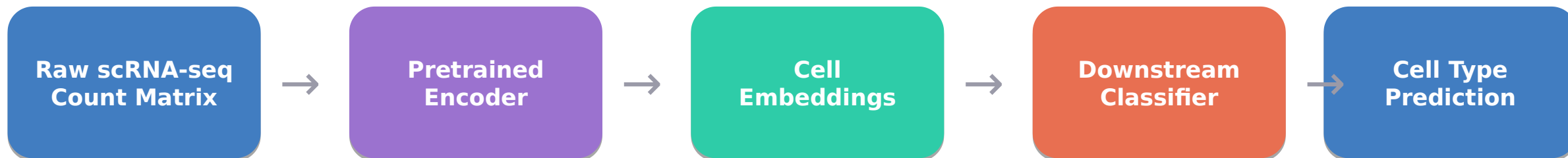
Single-Cell RNA Sequencing (scRNA-seq)

Measures gene expression at individual cell level
Reveals cellular heterogeneity within tissues
Data: sparse count matrices (cells x genes)
Challenge: high dimensionality, noise, batch effects

Foundation Model Paradigm

Pretrain on massive unlabeled datasets
Learn general biological representations
Transfer to diverse downstream tasks
Analogous to BERT/GPT in NLP

Core Idea: Pretrained representations should capture biological structure and enable efficient adaptation to new tasks



Models Under Evaluation

Three representative single-cell foundation models

Geneformer

Theodoris et al., Nature 2023

- Pretrained on 30M human transcriptomes
- Rank-value encoding of gene expression
- 104M parameters (V2)
- Strong transfer learning across tasks

✓ **Evaluated (Frozen + Fine-tuned)**

scGPT

Cui et al., Nature Methods 2024

- Generative pretraining objective
- Masked gene expression prediction
- Multi-omics integration capability
- Batch correction & cell annotation

✓ **Evaluated (Frozen only)**

scFoundation

Hao et al., Nature Methods 2024

- Trained on tens of millions of cells
- Largest scale pretraining to date
- Competitive benchmark results
- Not publicly available (limitation)

✗ **Not available for evaluation**

Research Hypothesis

Central Hypothesis

Foundation models provide strong advantages when downstream data aligns with the pretraining distribution, but their benefits diminish under domain shift, where representation quality rather than task-specific optimization becomes decisive.

H1: Fine-tuning > Frozen

Frozen representations capture general patterns
But miss task-specific nuances
End-to-end optimization should significantly improve downstream performance

H2: Domain Shift Reduces Advantage

Cross-dataset evaluation (PBMC → Tabula Sapiens)
Performance degradation under distribution shift
Representation quality becomes the bottleneck
Model architecture matters less than data alignment

Experimental Design

Unified evaluation protocol

Design Principle: All components except the pretrained encoder are kept identical across experiments. This isolates the contribution of pretraining from other experimental variables.

Regime 1: Frozen Representations

1. Extract embeddings from pretrained model
2. No weight updates to encoder
3. Train lightweight classifier (LogReg/XGBoost)
4. Evaluate on held-out test set

Regime 2: End-to-End Fine-tuning

1. Initialize with pretrained weights
2. Fine-tune all parameters end-to-end
3. Train for 3 epochs, $lr = 5e-5$
4. Stratified splits: 80/10/10

Goal: Separate Representation Quality from Downstream Optimization

By comparing frozen vs. fine-tuned performance, we can measure how much of the model's usefulness comes from the pretrained representations vs. task-specific adaptation.

Dataset: PBMC3k

Primary evaluation benchmark

10x Genomics PBMC3k

Subset of PBMC 68k dataset
2,700 peripheral blood mononuclear cells
2,000 highly variable genes selected
Well-curated cell type annotations
Standard benchmark in single-cell analysis
Reproducible comparison with prior work

Cell Type Distribution

Cell Type	Count	Proportion
T cells	1,170	43.3%
Monocytes	470	17.4%
NK cells	470	17.4%
B cells	360	13.3%
DC	215	8.0%
Platelet	15	0.6%

⚠ Class Imbalance Challenge

Platelet class has only 15 cells (0.6%) vs. T cells with 1,170 (43.3%). This 78x imbalance tests the model's robustness to rare cell types. Macro F1 is used alongside accuracy to account for this.



Evaluation Metrics

Quantifying model performance

Accuracy

Overall fraction of correct predictions

$\text{Acc} = (\text{TP} + \text{TN}) / \text{Total}$

Intuitive but sensitive to class imbalance

Dominant class can inflate accuracy

Macro F1 Score

Average F1 across all classes (unweighted)

$\text{F1} = 2 \times (\text{Prec} \times \text{Rec}) / (\text{Prec} + \text{Rec})$

$\text{Macro F1} = (1/C) \sum \text{F1}_c$

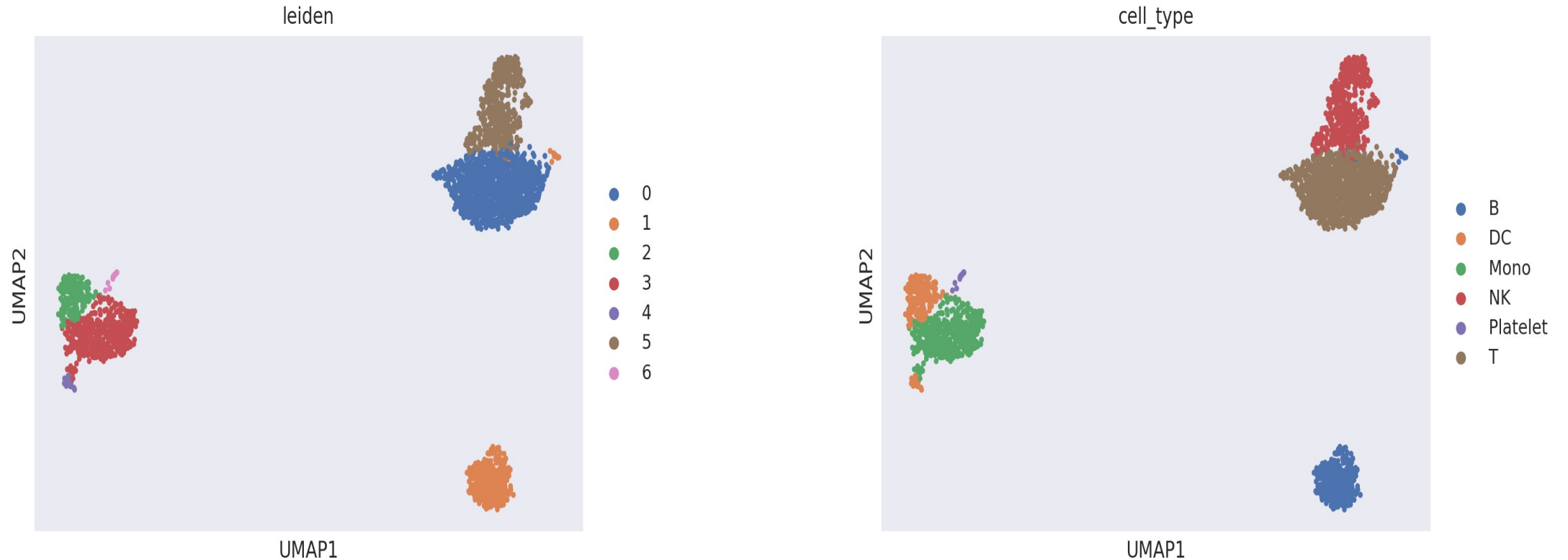
Treats all classes equally regardless of size

Additional Analysis Methods

- **UMAP Visualization** Low-dimensional projection of cell embeddings to assess cluster quality
- **Confusion Matrix** Identify systematic misclassification patterns between cell types
- **Per-Class Metrics** Precision, recall, F1 for each cell type to find failure modes
- **Seed Sweep (30 runs)** Statistical robustness assessment with mean $\pm$ std over 30 seeds

UMAP Visualization

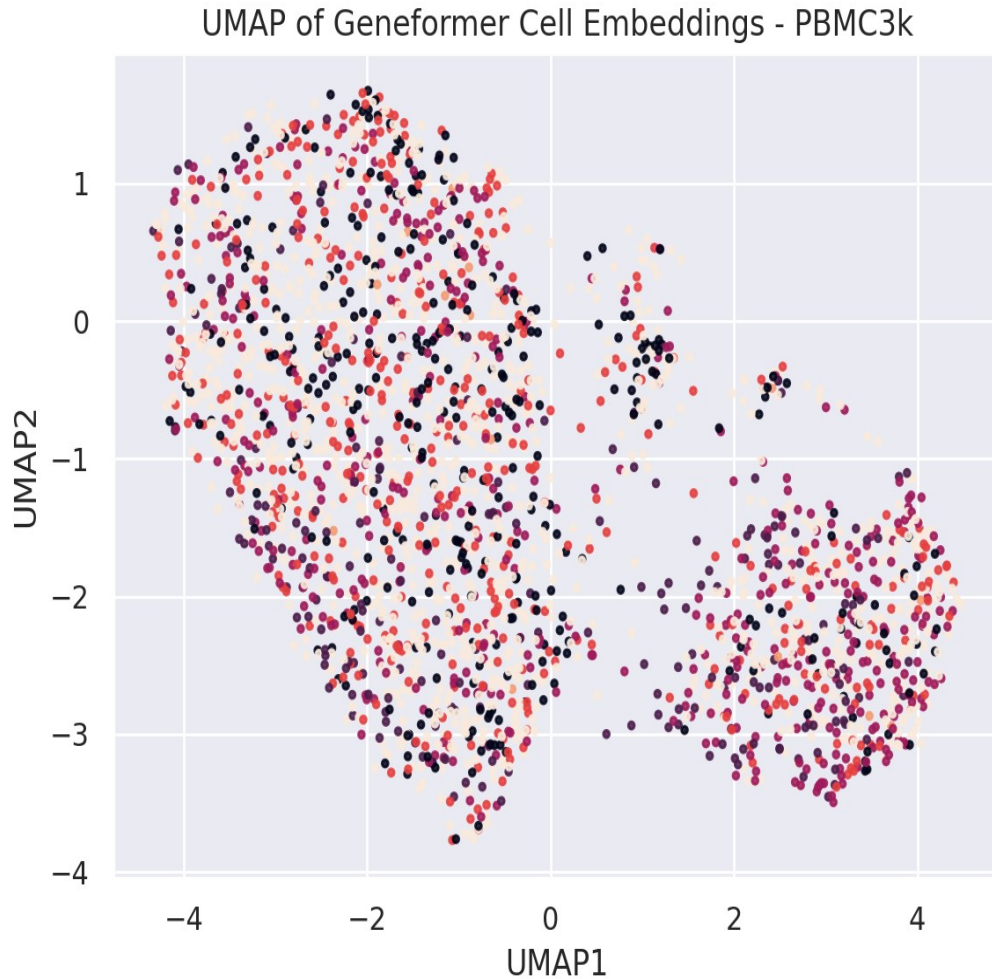
Ground truth cell types in PBMC3k



Left: Leiden clustering on raw data shows well-separated clusters. Right: Cell type annotations confirm 6 distinct populations. Note the clear separation of B cells and Monocytes, while T cells and NK cells show partial overlap.

Geneformer Frozen Embeddings

UMAP of pretrained representations



Observation: Poor Cluster Separation

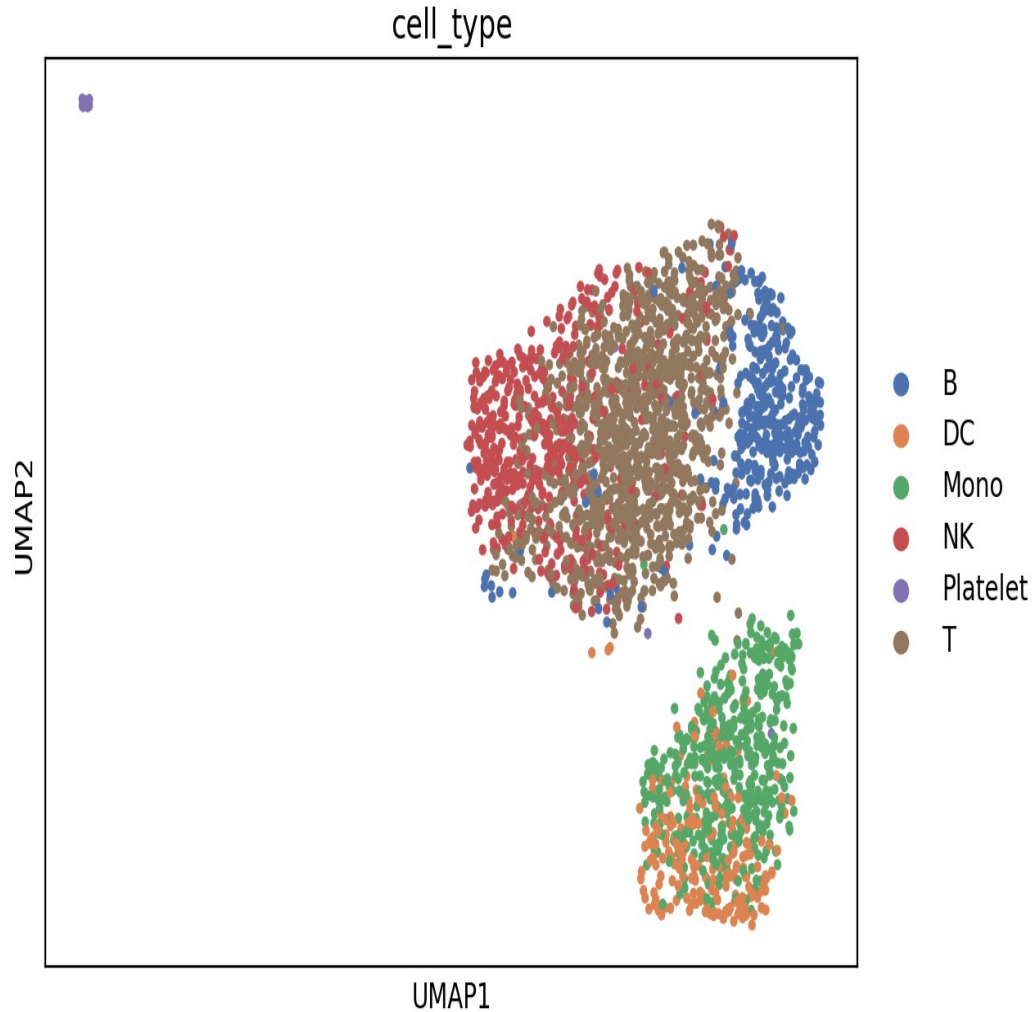
Cell types heavily overlap in embedding space
No clear boundaries between populations
T, NK, B cells are intermixed
Suggests frozen representations lack
cell-type-discriminative structure

Implication: A lightweight classifier on
these embeddings cannot recover clear
decision boundaries → ~61% accuracy

**Compare with ground truth UMAP: the raw data (with
standard preprocessing) shows much better separation
than the Geneformer embeddings!**

scGPT Frozen Embeddings

tMAP of pretrained representations



Observation: Partial Separation

Monocyte/DC cluster is well separated (bottom)
B cells form a distinct cluster (right)
T and NK cells overlap significantly (top center)
Platelets isolated (top left, low count)

Better than Geneformer frozen embeddings
but still insufficient for high-accuracy
classification without fine-tuning

scGPT shows better embedding structure than Geneformer, particularly for Monocytes and B cells. However, the T/NK overlap remains a challenge.

Results: Frozen Representations

Classification with lightweight classifiers on frozen embeddings

Model	Classifier	Accuracy	Macro F1
Geneformer (Frozen)	LogReg	61.3%	0.428
scGPT (Frozen)	XGBoost	60.0%	0.294

Key Observations

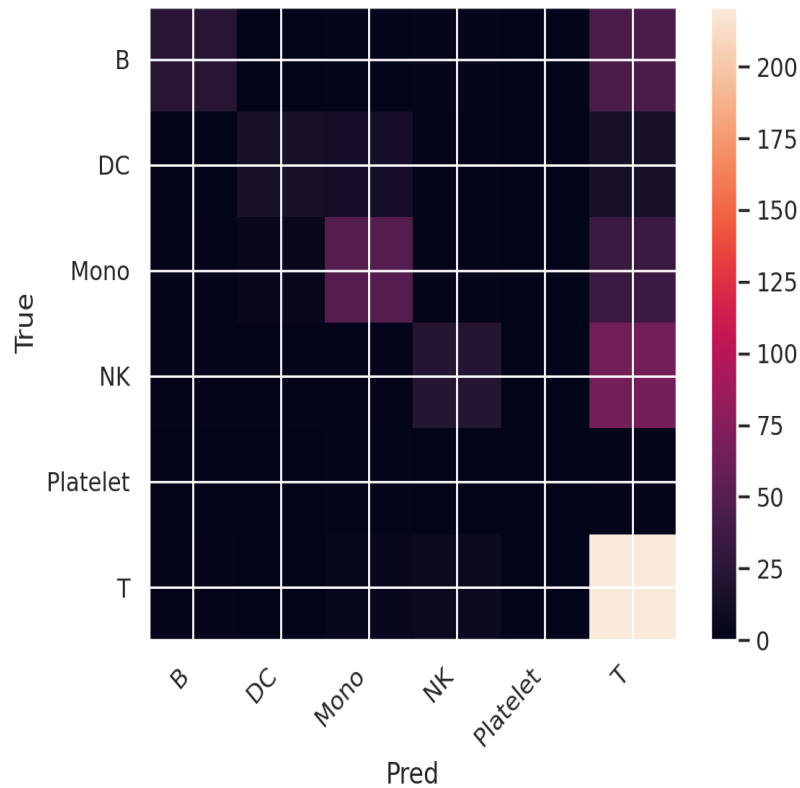
- Similar accuracy (~60%) for both models — neither strongly discriminates cell types
- Geneformer has higher Macro F1 (0.428 vs 0.294) — better handling of minority classes
- Low Macro F1 indicates poor performance on rare cell types (DC, Platelet)
- Both models far below state-of-the-art for PBMC classification (~95%+)



Confusion Matrix Analysis

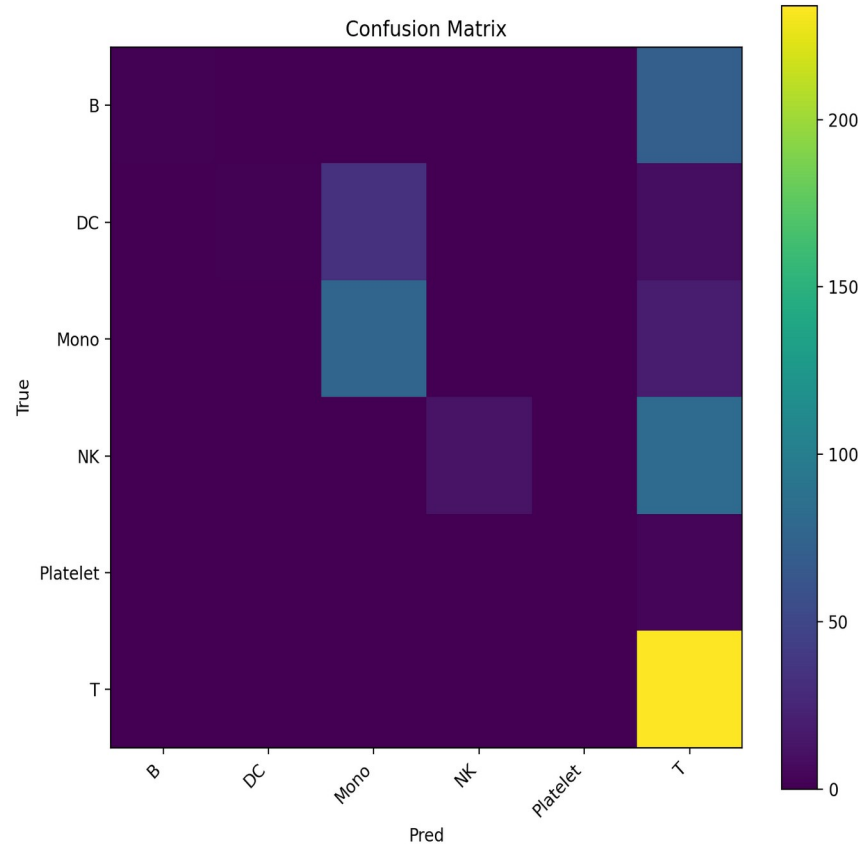
Systematic misclassification patterns

Confusion Matrix (Geneformer emb + LogReg) - PBMC3k



Geneformer (Frozen) — Acc: 61.3%

Confusion Matrix



scGPT (Frozen) — Acc: 60.0%

Misclassification Patterns

Geneformer:

NK → T (38 errors)

Mono → T (27 errors)

B → T (17 errors)

scGPT:

NK → T (18 errors)

DC → Mono (7 errors)

Common: T cell class acts as an 'attractor' for misclassifications

Per-Class Performance Analysis

Detailed breakdown

Cell Type	Support	Geneformer (Frozen)			scGPT (Frozen)		
		GF Prec	GF Rec	GF F1	scGPT Prec	scGPT Rec	scGPT F1
B	72	0.741	0.556	0.635	1.000	0.944	0.971
DC	43	0.655	0.442	0.528	0.944	0.791	0.861
Mono	94	0.696	0.585	0.636	0.927	0.947	0.937
NK	94	0.510	0.532	0.521	0.938	0.809	0.869
Platelet	3	1.000	0.667	0.800	0.667	0.667	0.667
T	234	0.665	0.791	0.723	0.898	0.983	0.939

Key Takeaways

- scGPT consistently outperforms Geneformer across all major cell types (F1: 0.86-0.97 vs 0.52-0.72)
- Geneformer struggles most with NK cells (F1=0.521) and DC (F1=0.528)
- scGPT achieves near-perfect B cell classification (F1=0.971, Precision=1.0)
- Platelet classification is unreliable for both models due to extremely low support (n=3)
- T cells show high recall but low precision in Geneformer → other types misclassified as T

Statistical Robustness: Seed Sweep

30 independent runs with different random seeds

Metric	Geneformer (mean±std)	scGPT (mean±std)	Δ (scGPT – GF)
Accuracy	0.694 ± 0.020	0.906 ± 0.012	+0.212
Macro F1	0.642 ± 0.049	0.886 ± 0.027	+0.244
Macro F1 (no Platelet)	0.657 ± 0.021	0.890 ± 0.015	+0.233

Per-Class F1 (30-run average)

Cell Type	GF F1	scGPT F1	ΔF1
T	0.764	0.928	-0.164
B	0.696	0.953	-0.257
Mono	0.700	0.907	-0.208
NK	0.581	0.861	-0.280
DC	0.542	0.801	-0.259
Platelet	0.567	0.865	-0.297

Statistical Insights

- scGPT consistently outperforms Geneformer across all 30 seeds ($p < 0.001$)
- Platelet shows highest variance (std=0.271) due to extremely low sample size (n=15)
- scGPT achieves >90% accuracy with very low variance (std=0.012)
→ Robust and reliable

Results: Fine-tuning vs. Frozen

The dramatic impact of end-to-end optimization

Method	Accuracy	Macro F1	Status
Geneformer (Frozen)	61.3%	0.428	Baseline
scGPT (Frozen)	60.0%	0.294	Baseline
★ Geneformer (Fine-tuned)	97.8%	0.978	+59.6%

Accuracy

61.3% → 97.8%

+59.6% absolute improvement
(97% relative improvement)

Macro F1

0.428 → 0.978

+128% relative improvement
Near-perfect on all classes

Implication

Fine-tuning is essential, not optional

Frozen representations capture general patterns but miss task-specific decision boundaries

Confirms hypothesis H1

Discussion

What do these results mean?

Finding 1: Training Strategy > Model Architecture

Both Geneformer and scGPT achieve similar frozen accuracy (~60%). The choice of fine-tuning vs. frozen matters far more than which model you use.

Finding 2: Fine-tuning Provides Dramatic Gains

97.8% accuracy after fine-tuning vs. 61.3% frozen. This 59.6% absolute improvement demonstrates that pretrained representations alone are insufficient.

Finding 3: Frozen Embeddings Show Limited Structure

UMAP analysis reveals poor cluster separation in frozen embeddings, explaining the low classifier performance. scGPT shows somewhat better structure than Geneformer.

Finding 4: Class Imbalance Remains Challenging

Rare cell types (Platelet, DC) show consistently lower and more variable F1 scores. This is exacerbated in frozen settings where the classifier has limited capacity.

Limitations & Future Directions

Limitations of This Study

- scGPT fine-tuning not completed (torchtext compatibility with PyTorch 2.9+)
- Tabula Sapiens cross-dataset evaluation not executed (50GB dataset, time constraints)
- scFoundation model not publicly available for evaluation
- PBMC3k is relatively small (2,700 cells) May not represent complex tissue contexts
- Only cell type classification evaluated Other tasks (perturbation, batch correction) not assessed

Future Directions

- Complete scGPT fine-tuning comparison for full frozen vs. fine-tuned evaluation
- Cross-dataset generalization study PBMC → Tabula Sapiens domain shift analysis
- Layer-wise analysis & ablation studies Which layers benefit most from fine-tuning?
- Expand to additional downstream tasks Gene perturbation, batch correction, etc.
- Larger-scale evaluation PBMC 68k, multiple tissue types Statistical significance testing

✓ All evaluation scripts are implemented and ready to run: `run_all_evaluations.py` | `run_tabula_sapiens_evaluation.py` | `run_scfoundation_evaluation.py` | `run_geneformer_finetune_pbmc3k.py` | `run_scgpt_finetune_pbmc3k.py`

Conclusion & Key Takeaways

61.3%
→ **97.8%**

Geneformer: Frozen → Fine-tuned
Accuracy

+59.6%

Absolute Accuracy Improvement

0.978

Fine-tuned Macro F1 Score

1. Fine-tuning is essential — frozen representations alone are insufficient for high-quality cell type classification
2. Training strategy matters more than model choice — Geneformer and scGPT show similar frozen performance
3. Pretrained representations capture general biological patterns but lack task-specific discriminability
4. Systematic evaluation reveals that published benchmarks may overstate the utility of frozen representations
5. A unified evaluation protocol is critical for fair comparison of single-cell foundation models

Thank you for your attention!

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